Hawaiʻi stress-test annex

Data as of 2026-10-28

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

Stress test, not a forecast. Each test changes one thing in the plan (a site closes, two hazards strike together, a road is cut, more people come) and counts who is affected, from the plan's own data. Deterministic code computes every number from the plan's data; people review the results.

People counted: residents of Census areas whose drive from the area's centre point to the nearest usable 2026 site was 15 minutes or less before the loss and is more than 15 minutes (or no route) after it; areas already beyond 15 minutes are not counted.

358 tests ran for Hawaiʻi; 99 affect someone or need county data and are listed below. Two-site losses are listed only when they add people beyond either site alone. This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

Hurricane: top 10

1. **Lose Kea‘au Armory and Pu‘u‘eo Community Center (hurricane)** (test 2: Two sites lost)
   48,669 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 9; people more than losing either site alone: 28,820).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 9 · Drive limit (minutes): 15 · Island: Hawai‘i · People more than losing either site alone: 28,820 · People left with no usable 2026 site on the island: 0 · Sites: Kea‘au Armory, Pu‘u‘eo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
2. **All four counties at once (hurricane)** (test 9: Several counties at once)
   43,517 people who may need a public shelter (likely), counties together (cots: 43,519; shelter staff per shift: 2,921).
   Cots: 43,519 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 247; Chuukese: 57; Hawaiian: 27; Ilocano: 369; Japanese: 201; Korean: 99; Marshallese: 24; Pidgin (Hawai‘i Creole English): 25; Spanish: 29; Tagalog/Filipino: 302; Vietnamese: 65 · Meals per day: 87,036 · People in shelter (likely): 43,517 · Shelter staff per shift: 2,921 · Water (gallons per day): 43,519
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).
3. **Honolulu and Hawai‘i at once (hurricane)** (test 9: Several counties at once)
   36,368 people who may need a public shelter (likely), counties together (cots: 36,368; shelter staff per shift: 2,433).
   Cots: 36,368 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 247; Chuukese: 57; Hawaiian: 23; Ilocano: 292; Japanese: 190; Korean: 99; Marshallese: 18; Pidgin (Hawai‘i Creole English): 25; Spanish: 15; Tagalog/Filipino: 260; Vietnamese: 65 · Meals per day: 72,736 · People in shelter (likely): 36,368 · Shelter staff per shift: 2,433 · Water (gallons per day): 36,368
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).
4. **Stranded visitors for 3 or 7 days – Hawai‘i** (test 21: Stranded visitors)
   35,640 visitors on an average day (island level).
   By length of stay: 3 days: 213,840 meals, 106,920 gallons of water; 7 days: 498,960 meals, 249,480 gallons of water · Visitors on an average day (2025): 35,640
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.

5. **Lose Kahilu Town Hall and Pu‘u‘eo Community Center (hurricane)** (test 2: Two sites lost)
   26,356 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 6,507).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 5 · Drive limit (minutes): 15 · Island: Hawai‘i · People more than losing either site alone: 6,507 · People left with no usable 2026 site on the island: 0 · Sites: Kahilu Town Hall, Pu‘u‘eo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
6. **Lose Pu‘u‘eo Community Center (hurricane)** (test 1: One site lost)
   19,849 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4).
   Census areas pushed outside a 15-minute drive: 4 · Drive limit (minutes): 15 · Island: Hawai‘i · May need a public shelter (likely): 595.5 · Site: Pu‘u‘eo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
7. **Lose Kea‘au Armory (hurricane)** (test 1: One site lost)
   13,194 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2).
   Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawai‘i · May need a public shelter (likely): 395.8 · Site: Kea‘au Armory
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
8. **Hurricane and inland flooding together – Hawai‘i** (test 5: Hurricane and inland flooding together)
   7,375 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 10; site gap: 20).
   Census areas: 43 · People if each hazard is counted separately: 267,681 · People in either footprint (counted once): 202,163 · May need a public shelter (likely): 7,375 · Site gap: 20 · Sites needed at 250 people per site (team assumption): 30 · 2026 sites usable for every hazard in the pair: 10
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

9. **Stays of 3 to 30 days – Hawai‘i (hurricane)** (test 11: Longer stays)
   6,065 people who may need a public shelter (likely).
   By length of stay: 14 days: 242,596 sq ft of floor space, 169,820 meals, 304 toilets, 84,910 gallons of water; 3 days: 121,298 sq ft of floor space, 36,390 meals, 152 toilets, 18,195 gallons of water; 30 days: 242,596 sq ft of floor space, 363,900 meals, 304 toilets, 181,950 gallons of water; 7 days: 242,596 sq ft of floor space, 84,910 meals, 304 toilets, 42,455 gallons of water · People in shelter (likely): 6,065 · Per person per day: 2 meals; 1 gallons of water
   Sources: Florida Division of Emergency Management, Sheltering Guidance Aid (2010); CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Floor space and toilets use the 2-week factors for stays over 3 days, also past 14 days: 40 sq ft; 1 toilet per 20.
10. **Māmalahoa Highway (Route 11) in Ka‘ū, between Pāhala and Punalu‘u** (test 24: Road closures (driving))
    5,591 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (Census areas cut off from every 2026 site: 0; most minutes added to any known site: 23.7).
    Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 1 · Census areas with no route at all: 0 · Census areas routed: 8 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 23.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 5,591 · Within reach means: 45-minute drive · Why this road: The only highway through Ka‘ū; lava, earthquake and flood exposure (team assumption).
    Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

Wildfire: top 10

1. **Stranded visitors for 3 or 7 days – Hawai‘i** (test 21: Stranded visitors)
   35,640 visitors on an average day (island level).
   By length of stay: 3 days: 213,840 meals, 106,920 gallons of water; 7 days: 498,960 meals, 249,480 gallons of water · Visitors on an average day (2025): 35,640
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **Areas high in the State Fire Risk layer but not in the plan's wildfire areas, and the reverse – Hawai‘i** (test 23b: State Fire Risk Areas vs the plan's wildfire areas)
   26,939 people in areas rated high only by the State Fire Risk layer (Census areas high in the State layer only: 7; Census areas in the plan's wildfire areas only: 7).
   Areas high in both, of all areas high in either: 22% · Census areas high in both: 4 · Census areas in the plan's wildfire areas only: 7 · Census areas high in the State layer only: 7 · Largest plan-only areas: Hawaiian Paradise Park · Census area 210.14, Hāwī / Kapa‘au · Census area 218, Hōlualoa · Census area 215.02, Hawaiian Acres / Fern Acres · Census area 210.16, Kurtistown · Census area 210.13 · People in plan-only areas: 37,926 · People in State-only areas: 26,939 · People in the plan's wildfire areas (FEMA National Risk Index): 57,557 · People in State High fire risk areas: 47,251 · Largest State-only areas: Kailua · Census area 215.04, Kailua · Census area 216.07, Kailua · Census area 216.08, Kailua · Census area 216.05, Kailua · Census area 216.06
   Sources: State Fire Risk Areas (DLNR Division of Forestry and Wildlife, via Hawai‘i Statewide GIS); FEMA National Risk Index wildfire scores; CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks. Team assumptions: An area is 'high' in the State layer when at least half its people live in a High fire risk area: 50%.
3. **All four counties at once (wildfire)** (test 9: Several counties at once)
   22,115 people who may need a public shelter (likely), counties together (cots: 22,116; shelter staff per shift: 1,534).
   Cots: 22,116 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 61; Chuukese: 14; Hawaiian: 9; Ilocano: 110; Japanese: 53; Korean: 24; Marshallese: 7; Pidgin (Hawai‘i Creole English): 9; Spanish: 12; Tagalog/Filipino: 82; Vietnamese: 16 · Meals per day: 44,231 · People in shelter (likely): 22,115 · Shelter staff per shift: 1,534 · Water (gallons per day): 22,116
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).

4. **Lose Kahilu Town Hall and Kea‘au Armory (wildfire)** (test 2: Two sites lost)
   19,701 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 6,507).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Hawai‘i · People more than losing either site alone: 6,507 · People left with no usable 2026 site on the island: 0 · Sites: Kahilu Town Hall, Kea‘au Armory
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
5. **Honolulu and Hawai‘i at once (wildfire)** (test 9: Several counties at once)
   18,591 people who may need a public shelter (likely), counties together (cots: 18,592; shelter staff per shift: 1,285).
   Cots: 18,592 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 61; Chuukese: 14; Hawaiian: 7; Ilocano: 77; Japanese: 48; Korean: 24; Marshallese: 5; Pidgin (Hawai‘i Creole English): 9; Spanish: 6; Tagalog/Filipino: 64; Vietnamese: 16 · Meals per day: 37,183 · People in shelter (likely): 18,591 · Shelter staff per shift: 1,285 · Water (gallons per day): 18,592
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).
6. **Lose Kea‘au Armory (wildfire)** (test 1: One site lost)
   13,194 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2).
   Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawai‘i · May need a public shelter (likely): 659.7 · Site: Kea‘au Armory
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
7. **Lose Honoka‘a Gym and Kea‘au Armory (wildfire)** (test 2: Two sites lost)
   13,194 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 0).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawai‘i · People more than losing either site alone: 0 · People left with no usable 2026 site on the island: 0 · Sites: Honoka‘a Gym, Kea‘au Armory
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

8. **Lose Kahilu Town Hall (wildfire)** (test 1: One site lost)
   6,507 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1).
   Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Hawai‘i · May need a public shelter (likely): 325.4 · Site: Kahilu Town Hall
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
9. **Māmalahoa Highway (Route 11) in Ka‘ū, between Pāhala and Punalu‘u** (test 24: Road closures (driving))
   5,591 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (Census areas cut off from every 2026 site: 0; most minutes added to any known site: 23.7).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 1 · Census areas with no route at all: 0 · Census areas routed: 8 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 23.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 5,591 · Within reach means: 45-minute drive · Why this road: The only highway through Ka‘ū; lava, earthquake and flood exposure (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.
10. **Wildfire and power loss together – Hawai‘i** (test 6: Wildfire and power loss together)
    2,878 people who may need a public shelter (likely) (outlets per day if the power is out: 421).
    Census areas: 11 · People if each hazard is counted separately: 57,557 · Outlets per day if the power is out: 421 · Outlets in the plan (people in shelter): 22 · People in either footprint (counted once): 57,557 · Power-dependent residents in the risk area: 421 · Power outage area: needs county data (utility outage areas) · May need a public shelter (likely): 2,878 · Backup power at sites: needs county data · Site gap: 2 · Sites needed at 250 people per site (team assumption): 12 · 2026 sites usable for every hazard in the pair: 10
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate; Power is out across the wildfire risk area; every power-dependent resident needs one charge a day: 1.

Tsunami: top 10

1. **Visitors on the island(s) – Hawai‘i (tsunami)** (test 20: Visitors in tsunami zones)
   35,640 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations)).
   Detail: Hawai‘i Island · Peak day: needs DBEDT monthly data (the Data Book gives the annual average) · Visitors on an average day (2025): 35,640 · Visitors inside the tsunami zone: needs county data (hotel and rental locations)
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025).

2. **Stranded visitors for 3 or 7 days – Hawai‘i** (test 21: Stranded visitors)
   35,640 visitors on an average day (island level).
   By length of stay: 3 days: 213,840 meals, 106,920 gallons of water; 7 days: 498,960 meals, 249,480 gallons of water · Visitors on an average day (2025): 35,640
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.

3. **Extreme tsunami zone (worst case) – Hawai‘i** (test 8: Extreme tsunami zone (worst case))
   13,808 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 690; 2026 sites inside the extreme zone: 0).
   People in the extreme tsunami zone: 13,808 · People in the standard tsunami zone: 12,102 · May need a shelter (high): 1,519 · May need a public shelter (likely): 690 · May need a shelter (low): 276 · 2026 sites inside the extreme zone: 0 · Sites needed at 250 people per site (team assumption): 3 · 2026 sites usable outside both zones: 10
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; State tsunami evacuation zones and extreme tsunami evacuation zones (Hawai‘i Statewide GIS); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250.

4. **All four counties at once (tsunami)** (test 9: Several counties at once)
   6,382 people who may need a public shelter (likely), counties together (cots: 6,384; shelter staff per shift: 442).
   Cots: 6,384 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 31; Chuukese: 7; Hawaiian: 4; Ilocano: 60; Japanese: 28; Korean: 12; Marshallese: 3; Pidgin (Hawai‘i Creole English): 3; Spanish: 6; Tagalog/Filipino: 45; Vietnamese: 8 · Meals per day: 12,766 · People in shelter (likely): 6,382 · Shelter staff per shift: 442 · Water (gallons per day): 6,384
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).

5. **Māmalahoa Highway (Route 11) in Kaʻū, between Pāhala and Punaluʻu** (test 24: Road closures (driving))
   5,591 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (Census areas cut off from every 2026 site: 0; most minutes added to any known site: 23.7).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 1 · Census areas with no route at all: 0 · Census areas routed: 8 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 23.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 5,591 · Within reach means: 45-minute drive · Why this road: The only highway through Kaʻū; lava, earthquake and flood exposure (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawaiʻi DOT HPMS and surveyed mile markers (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Hawaiʻi DOE public schools (Hawaiʻi Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

6. **Honolulu and Hawaiʻi at once (tsunami)** (test 9: Several counties at once)
   4,622 people who may need a public shelter (likely), counties together (cots: 4,624; shelter staff per shift: 316).
   Cots: 4,624 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 31; Chuukese: 7; Hawaiian: 3; Ilocano: 37; Japanese: 24; Korean: 12; Marshallese: 2; Pidgin (Hawaiʻi Creole English): 3; Spanish: 2; Tagalog/Filipino: 32; Vietnamese: 8 · Meals per day: 9,246 · People in shelter (likely): 4,622 · Shelter staff per shift: 316 · Water (gallons per day): 4,624
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).

7. **Lose Keaʻau Armory and Puʻuʻeo Community Center (tsunami)** (test 2: Two sites lost)
   3,604 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 2,125).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 4 · Drive limit (minutes): 15 · Island: Hawaiʻi · People more than losing either site alone: 2,125 · People left with no usable 2026 site on the island: 0 · Sites: Keaʻau Armory, Puʻuʻeo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

8. **Tsunami and coastal flooding together – Hawai‘i** (test 4: Tsunami and coastal flooding together)
   2,771 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 10; site gap: 2).
   Census areas: 17 · People if each hazard is counted separately: 66,071 · People in either footprint (counted once): 55,427 · May need a public shelter (likely): 2,771 · Site gap: 2 · Sites needed at 250 people per site (team assumption): 12 · 2026 sites usable for every hazard in the pair: 10
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions); State tsunami evacuation zones and extreme tsunami evacuation zones (Hawai‘i Statewide GIS). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.
9. **Lose Pu‘u‘eo Community Center (tsunami)** (test 1: One site lost)
   1,479 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2).
   Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawai‘i · May need a public shelter (likely): 74 · Site: Pu‘u‘eo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
10. **Lose Honoka‘a Gym and Pu‘u‘eo Community Center (tsunami)** (test 2: Two sites lost)
    1,479 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 0).
    Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawai‘i · People more than losing either site alone: 0 · People left with no usable 2026 site on the island: 0 · Sites: Honoka‘a Gym, Pu‘u‘eo Community Center
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

Inland flooding: top 10

1. **Stranded visitors for 3 or 7 days – Hawai‘i** (test 21: Stranded visitors)
   35,640 visitors on an average day (island level).
   By length of stay: 3 days: 213,840 meals, 106,920 gallons of water; 7 days: 498,960 meals, 249,480 gallons of water · Visitors on an average day (2025): 35,640
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **Lose Kea‘au Armory and Pu‘u‘eo Community Center (inland flooding)** (test 2: Two sites lost)
   35,169 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 6; people more than losing either site alone: 21,975).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 6 · Drive limit (minutes): 15 · Island: Hawai‘i · People more than losing either site alone: 21,975 · People left with no usable 2026 site on the island: 0 · Sites: Kea‘au Armory, Pu‘u‘eo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
3. **All four counties at once (inland flooding)** (test 9: Several counties at once)
   25,452 people who may need a public shelter (likely), counties together (cots: 25,454; shelter staff per shift: 1,723).
   Cots: 25,454 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 100; Chuukese: 23; Hawaiian: 11; Ilocano: 152; Japanese: 82; Korean: 40; Marshallese: 11; Pidgin (Hawai‘i Creole English): 11; Spanish: 13; Tagalog/Filipino: 123; Vietnamese: 26 · Meals per day: 50,906 · People in shelter (likely): 25,452 · Shelter staff per shift: 1,723 · Water (gallons per day): 25,454
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).
4. **Honolulu and Hawai‘i at once (inland flooding)** (test 9: Several counties at once)
   21,749 people who may need a public shelter (likely), counties together (cots: 21,750; shelter staff per shift: 1,468).
   Cots: 21,750 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 100; Chuukese: 23; Hawaiian: 9; Ilocano: 120; Japanese: 77; Korean: 40; Marshallese: 8; Pidgin (Hawai‘i Creole English): 11; Spanish: 7; Tagalog/Filipino: 105; Vietnamese: 26 · Meals per day: 43,499 · People in shelter (likely): 21,749 · Shelter staff per shift: 1,468 · Water (gallons per day): 21,750
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).

5. **Lose Kahilu Town Hall and Kea‘au Armory (inland flooding)** (test 2: Two sites lost)
   19,701 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 6,507).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Hawai‘i · People more than losing either site alone: 6,507 · People left with no usable 2026 site on the island: 0 · Sites: Kahilu Town Hall, Kea‘au Armory
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
6. **Lose Kea‘au Armory (inland flooding)** (test 1: One site lost)
   13,194 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2).
   Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawai‘i · May need a public shelter (likely): 659.7 · Site: Kea‘au Armory
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
7. **Lose Pu‘u‘eo Community Center (inland flooding)** (test 1: One site lost)
   11,177 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2).
   Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawai‘i · May need a public shelter (likely): 558.9 · Site: Pu‘u‘eo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
8. **Hurricane and inland flooding together – Hawai‘i** (test 5: Hurricane and inland flooding together)
   7,375 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 10; site gap: 20).
   Census areas: 43 · People if each hazard is counted separately: 267,681 · People in either footprint (counted once): 202,163 · May need a public shelter (likely): 7,375 · Site gap: 20 · Sites needed at 250 people per site (team assumption): 30 · 2026 sites usable for every hazard in the pair: 10
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

9. **Māmalahoa Highway (Route 11) in Kaʻū, between Pāhala and Punaluʻu** (test 24: Road closures (driving))

   5,591 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (Census areas cut off from every 2026 site: 0; most minutes added to any known site: 23.7).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 1 · Census areas with no route at all: 0 · Census areas routed: 8 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 23.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 5,591 · Within reach means: 45-minute drive · Why this road: The only highway through Kaʻū; lava, earthquake and flood exposure (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawaiʻi DOT HPMS and surveyed mile markers (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Hawaiʻi DOE public schools (Hawaiʻi Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

10. **Dam failure and inland flooding together – Hawaiʻi** (test 7: Dam failure and inland flooding together)

    3,412 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 9; site gap: 5).

    Census areas: 14 · People if each hazard is counted separately: 70,063 · People in either footprint (counted once): 68,239 · May need a public shelter (likely): 3,412 · Site gap: 5 · Sites needed at 250 people per site (team assumption): 14 · 2026 sites usable for every hazard in the pair: 9

    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions); State of Hawaiʻi dam evacuation zones (Hawaiʻi Statewide GIS). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate; Dam failure uses the inland flooding shelter-seeking rate: 5%.

Coastal flooding: top 10

1. **Stranded visitors for 3 or 7 days – Hawai‘i** (test 21: Stranded visitors)
   35,640 visitors on an average day (island level).
   By length of stay: 3 days: 213,840 meals, 106,920 gallons of water; 7 days: 498,960 meals, 249,480 gallons of water · Visitors on an average day (2025): 35,640
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **All four counties at once (coastal flooding)** (test 9: Several counties at once)
   17,661 people who may need a public shelter (likely), counties together (cots: 17,664; shelter staff per shift: 1,171).
   Cots: 17,664 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 98; Chuukese: 22; Hawaiian: 12; Ilocano: 153; Japanese: 82; Korean: 39; Marshallese: 11; Pidgin (Hawai‘i Creole English): 14; Spanish: 14; Tagalog/Filipino: 122; Vietnamese: 26 · Meals per day: 35,325 · People in shelter (likely): 17,661 · Shelter staff per shift: 1,171 · Water (gallons per day): 17,664
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).
3. **Honolulu and Hawai‘i at once (coastal flooding)** (test 9: Several counties at once)
   14,868 people who may need a public shelter (likely), counties together (cots: 14,870; shelter staff per shift: 977).
   Cots: 14,870 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 98; Chuukese: 22; Hawaiian: 10; Ilocano: 121; Japanese: 78; Korean: 39; Marshallese: 9; Pidgin (Hawai‘i Creole English): 14; Spanish: 8; Tagalog/Filipino: 104; Vietnamese: 26 · Meals per day: 29,738 · People in shelter (likely): 14,868 · Shelter staff per shift: 977 · Water (gallons per day): 14,870
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).
4. **Lose Kea‘au Armory and Pu‘u‘eo Community Center (coastal flooding)** (test 2: Two sites lost)
   13,117 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 5,552).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 3 · Drive limit (minutes): 15 · Island: Hawai‘i · People more than losing either site alone: 5,552 · People left with no usable 2026 site on the island: 0 · Sites: Kea‘au Armory, Pu‘u‘eo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

5. **Lose Puʻuʻeo Community Center (coastal flooding)** (test 1: One site lost)
   7,565 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2).
   Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawaiʻi · May need a public shelter (likely): 378.2 · Site: Puʻuʻeo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
6. **Lose Honokaʻa Gym and Puʻuʻeo Community Center (coastal flooding)** (test 2: Two sites lost)
   7,565 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 0).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Hawaiʻi · People more than losing either site alone: 0 · People left with no usable 2026 site on the island: 0 · Sites: Honokaʻa Gym, Puʻuʻeo Community Center
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
7. **Māmalahoa Highway (Route 11) in Kaʻū, between Pāhala and Punaluʻu** (test 24: Road closures (driving))
   5,591 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (Census areas cut off from every 2026 site: 0; most minutes added to any known site: 23.7).
   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 1 · Census areas with no route at all: 0 · Census areas routed: 8 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 23.7 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 5,591 · Within reach means: 45-minute drive · Why this road: The only highway through Kaʻū; lava, earthquake and flood exposure (team assumption).
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawaiʻi DOT HPMS and surveyed mile markers (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Hawaiʻi DOE public schools (Hawaiʻi Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

8. **Tsunami and coastal flooding together – Hawai‘i** (test 4: Tsunami and coastal flooding together)
   2,771 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 10; site gap: 2).
   Census areas: 17 · People if each hazard is counted separately: 66,071 · People in either footprint (counted once): 55,427 · May need a public shelter (likely): 2,771 · Site gap: 2 · Sites needed at 250 people per site (team assumption): 12 · 2026 sites usable for every hazard in the pair: 10
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions); State tsunami evacuation zones and extreme tsunami evacuation zones (Hawai‘i Statewide GIS). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.
9. **Rate at which demand needs more sites than are usable – Hawai‘i (coastal flooding)** (test 10: Shelter-seeking rate higher than planned)
   2,698 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 5%; usable 2026 sites: 10).
   Known 2026 sites in the county: 10 · Not reached by: – · People in the risk area: 53,969 · Likely rate in the plan: 5% · Rate when usable sites run out: 5% · Sites needed by rate (250 people per site): 10%: 22; 11%: 24; 12%: 26; 13%: 29; 14%: 31; 15%: 33; 16%: 35; 17%: 37; 18%: 39; 19%: 42; 2%: 5; 20%: 44; 3%: 7; 4%: 9; 5%: 11; 6%: 13; 7%: 16; 8%: 18; 9%: 20 · Usable 2026 sites: 10
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; Rates tested: 2–20% in 1-point steps.
10. **Stays of 3 to 30 days – Hawai‘i (coastal flooding)** (test 11: Longer stays)
    2,698 people who may need a public shelter (likely).
    By length of stay: 14 days: 107,939 sq ft of floor space, 75,558 meals, 135 toilets, 37,786 gallons of water; 3 days: 53,970 sq ft of floor space, 16,191 meals, 68 toilets, 8,097 gallons of water; 30 days: 107,939 sq ft of floor space, 161,910 meals, 135 toilets, 80,970 gallons of water; 7 days: 107,939 sq ft of floor space, 37,779 meals, 135 toilets, 18,893 gallons of water · People in shelter (likely): 2,698 · Per person per day: 2 meals; 1 gallons of water
    Sources: Florida Division of Emergency Management, Sheltering Guidance Aid (2010); CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Floor space and toilets use the 2-week factors for stays over 3 days, also past 14 days: 40 sq ft; 1 toilet per 20.

Every result, ranked by people affected

Rank	Test	Scenario	Hazard	Result
1	2	Lose Kea‘au Armory and Pu‘u‘eo Community Center (hurricane)	Hurricane	48,669 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 9; people more than losing either site alone: 28,820)
1	9	All four counties at once (hurricane)	Hurricane	43,517 people who may need a public shelter (likely), counties together (cots: 43,519; shelter staff per shift: 2,921)
2	9	Honolulu and Hawai‘i at once (hurricane)	Hurricane	36,368 people who may need a public shelter (likely), counties together (cots: 36,368; shelter staff per shift: 2,433)
2	20	Visitors on the island(s) – Hawai‘i (tsunami)	Tsunami	35,640 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations))
3	21	Stranded visitors for 3 or 7 days – Hawai‘i	All hazards	35,640 visitors on an average day (island level)
4	2	Lose Kea‘au Armory and Pu‘u‘eo Community Center (inland flooding)	Inland flooding	35,169 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 6; people more than losing either site alone: 21,975)
5	23b	Areas high in the State Fire Risk layer but not in the plan's wildfire areas, and the reverse – Hawai‘i	Wildfire	26,939 people in areas rated high only by the State Fire Risk layer (Census areas high in the State layer only: 7; Census areas in the plan's wildfire areas only: 7)
5	9	All four counties at once (inland flooding)	Inland flooding	25,452 people who may need a public shelter (likely), counties together (cots: 25,454; shelter staff per shift: 1,723)
6	2	Lose Kahilu Town Hall and Pu‘u‘eo Community Center (hurricane)	Hurricane	26,356 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 6,507)

Rank	Test	Scenario	Hazard	Result
6	9	All four counties at once (wildfire)	Wildfire	22,115 people who may need a public shelter (likely), counties together (cots: 22,116; shelter staff per shift: 1,534)
7	2	Lose Puʻuʻeo Community Center and Yano Hall (hurricane)	Hurricane	23,707 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 3,858)
7	9	Honolulu and Hawaiʻi at once (inland flooding)	Inland flooding	21,749 people who may need a public shelter (likely), counties together (cots: 21,750; shelter staff per shift: 1,468)
8	1	Lose Puʻuʻeo Community Center (hurricane)	Hurricane	19,849 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4)
10	9	Honolulu and Hawaiʻi at once (wildfire)	Wildfire	18,591 people who may need a public shelter (likely), counties together (cots: 18,592; shelter staff per shift: 1,285)
12	9	All four counties at once (coastal flooding)	Coastal flooding	17,661 people who may need a public shelter (likely), counties together (cots: 17,664; shelter staff per shift: 1,171)
14	9	Honolulu and Hawaiʻi at once (coastal flooding)	Coastal flooding	14,868 people who may need a public shelter (likely), counties together (cots: 14,870; shelter staff per shift: 977)
15	2	Lose Kahilu Town Hall and Keaʻau Armory (hurricane)	Hurricane	19,701 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 6,507)
16	2	Lose Kahilu Town Hall and Keaʻau Armory (inland flooding)	Inland flooding	19,701 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 6,507)

Rank	Test	Scenario	Hazard	Result
17	2	Lose Kahilu Town Hall and Keaʻau Armory (wildfire)	Wildfire	19,701 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 6,507)
17	9	Maui and Hawaiʻi at once (hurricane)	Hurricane	11,009 people who may need a public shelter (likely), counties together (cots: 11,010; shelter staff per shift: 744)
18	2	Lose Kahilu Town Hall and Puʻuʻeo Community Center (inland flooding)	Inland flooding	17,684 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 6,507)
18	9	Kauaʻi and Hawaiʻi at once (hurricane)	Hurricane	8,270 people who may need a public shelter (likely), counties together (cots: 8,271; shelter staff per shift: 564)
19	2	Lose Keaʻau Armory and Yano Hall (hurricane)	Hurricane	17,052 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 3,858)
20	8	Extreme tsunami zone (worst case) – Hawaiʻi	Tsunami	13,808 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 690; 2026 sites inside the extreme zone: 0)
20	9	All four counties at once (tsunami)	Tsunami	6,382 people who may need a public shelter (likely), counties together (cots: 6,384; shelter staff per shift: 442)
21	1	Lose Keaʻau Armory (hurricane)	Hurricane	13,194 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)
21	9	Maui and Hawaiʻi at once (inland flooding)	Inland flooding	5,909 people who may need a public shelter (likely), counties together (cots: 5,909; shelter staff per shift: 401)
22	1	Lose Keaʻau Armory (inland flooding)	Inland flooding	13,194 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)

Rank	Test	Scenario	Hazard	Result
23	1	Lose Kea‘au Armory (wildfire)	Wildfire	13,194 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)
23	9	Maui and Hawai‘i at once (wildfire)	Wildfire	5,320 people who may need a public shelter (likely), counties together (cots: 5,320; shelter staff per shift: 371)
24	9	Honolulu and Hawai‘i at once (tsunami)	Tsunami	4,622 people who may need a public shelter (likely), counties together (cots: 4,624; shelter staff per shift: 316)
25	9	Maui and Hawai‘i at once (coastal flooding)	Coastal flooding	4,532 people who may need a public shelter (likely), counties together (cots: 4,533; shelter staff per shift: 303)
27	9	Kaua‘i and Hawai‘i at once (inland flooding)	Inland flooding	4,346 people who may need a public shelter (likely), counties together (cots: 4,347; shelter staff per shift: 300)
28	9	Kaua‘i and Hawai‘i at once (wildfire)	Wildfire	3,960 people who may need a public shelter (likely), counties together (cots: 3,960; shelter staff per shift: 276)
30	9	Kaua‘i and Hawai‘i at once (coastal flooding)	Coastal flooding	3,657 people who may need a public shelter (likely), counties together (cots: 3,659; shelter staff per shift: 245)
33	9	Maui and Hawai‘i at once (tsunami)	Tsunami	1,913 people who may need a public shelter (likely), counties together (cots: 1,914; shelter staff per shift: 136)
35	9	Kaua‘i and Hawai‘i at once (tsunami)	Tsunami	1,057 people who may need a public shelter (likely), counties together (cots: 1,058; shelter staff per shift: 80)
45	2	Lose Kea‘au Armory and Pu‘u‘eo Community Center (coastal flooding)	Coastal flooding	13,117 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 5,552)
46	1	Lose Pu‘u‘eo Community Center (inland flooding)	Inland flooding	11,177 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)

Rank	Test	Scenario	Hazard	Result
54	2	Lose Kahilu Town Hall and Yano Hall (hurricane)	Hurricane	10,365 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 3,858)
55	1	Lose Puʻuʻeo Community Center (coastal flooding)	Coastal flooding	7,565 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)
64	5	Hurricane and inland flooding together – Hawaiʻi	Hurricane + Inland flooding	7,375 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 10; site gap: 20)
65	1	Lose Kahilu Town Hall (hurricane)	Hurricane	6,507 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
66	1	Lose Kahilu Town Hall (inland flooding)	Inland flooding	6,507 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
67	1	Lose Kahilu Town Hall (wildfire)	Wildfire	6,507 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
89	11	Stays of 3 to 30 days – Hawaiʻi (hurricane)	Hurricane	6,065 people who may need a public shelter (likely)
90	24	Māmalahoa Highway (Route 11) in Kaʻū, between Pāhala and Punaluʻu	All hazards	5,591 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (Census areas cut off from every 2026 site: 0; most minutes added to any known site: 23.7)
91	13	Kūpuna living alone, no car, far from a site – Hawaiʻi (hurricane)	Hurricane	4,258 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 32)
92	10	Rate at which demand needs more sites than are usable – Hawaiʻi (hurricane)	Hurricane	4,043 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 2%; usable 2026 sites: 10)

Rank	Test	Scenario	Hazard	Result
93	1	Lose Yano Hall (hurricane)	Hurricane	3,858 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
100	2	Lose Kea‘au Armory and Pu‘u‘eo Community Center (tsunami)	Tsunami	3,604 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 2,125)
101	7	Dam failure and inland flooding together – Hawai‘i	Dam failure + Inland flooding	3,412 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 9; site gap: 5)
102	11	Stays of 3 to 30 days – Hawai‘i (inland flooding)	Inland flooding	3,276 people who may need a public shelter (likely)
103	6	Wildfire and power loss together – Hawai‘i	Wildfire + Power loss	2,878 people who may need a public shelter (likely) (outlets per day if the power is out: 421)
104	10	Rate at which demand needs more sites than are usable – Hawai‘i (wildfire)	Wildfire	2,878 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 5%; usable 2026 sites: 10)
105	11	Stays of 3 to 30 days – Hawai‘i (wildfire)	Wildfire	2,878 people who may need a public shelter (likely)
106	4	Tsunami and coastal flooding together – Hawai‘i	Tsunami + Coastal flooding	2,771 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 10; site gap: 2)
107	10	Rate at which demand needs more sites than are usable – Hawai‘i (coastal flooding)	Coastal flooding	2,698 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 5%; usable 2026 sites: 10)
108	11	Stays of 3 to 30 days – Hawai‘i (coastal flooding)	Coastal flooding	2,698 people who may need a public shelter (likely)
109	10	Rate at which demand needs more sites than are usable – Hawai‘i (inland flooding)	Inland flooding	2,621 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 4%; usable 2026 sites: 10)
110	1	Lose Pu‘u‘eo Community Center (tsunami)	Tsunami	1,479 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)

Rank	Test	Scenario	Hazard	Result
119	12	Power out 1 to 14 days – Hawai‘i (hurricane)	Hurricane	1,450 power-dependent residents in the risk area (outlets per day while the power is out: 1,450)
120	13	Kūpuna living alone, no car, far from a site – Hawai‘i (coastal flooding)	Coastal flooding	1,124 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 8)
121	15	Power-dependent and far from a site – Hawai‘i (hurricane)	Hurricane	1,018 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 32)
122	13	Kūpuna living alone, no car, far from a site – Hawai‘i (inland flooding)	Inland flooding	960 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 4)
123	1	Lose Kea‘au Armory (tsunami)	Tsunami	765 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
132	13	Kūpuna living alone, no car, far from a site – Hawai‘i (wildfire)	Wildfire	755 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 8)
133	11	Stays of 3 to 30 days – Hawai‘i (tsunami)	Tsunami	605 people who may need a public shelter (likely)
134	16	Marshallese: reliable count, 0 of 15 messages translated – Hawai‘i	All hazards	538 people who speak English less than very well (messages translated: 0)
135	12	Power out 1 to 14 days – Hawai‘i (inland flooding)	Inland flooding	484 power-dependent residents in the risk area (outlets per day while the power is out: 484)
136	18	Add the 2026 Point-in-Time count – Hawai‘i (coastal flooding)	Coastal flooding	467 people added by the 2026 Point-in-Time count (increase over the likely demand: 17.3%)
137	18	Add the 2026 Point-in-Time count – Hawai‘i (hurricane)	Hurricane	467 people added by the 2026 Point-in-Time count (increase over the likely demand: 7.7%)
138	18	Add the 2026 Point-in-Time count – Hawai‘i (inland flooding)	Inland flooding	467 people added by the 2026 Point-in-Time count (increase over the likely demand: 14.3%)

Rank	Test	Scenario	Hazard	Result
139	18	Add the 2026 Point-in-Time count – Hawai‘i (tsunami)	Tsunami	467 people added by the 2026 Point-in-Time count (increase over the likely demand: 77.2%)
140	18	Add the 2026 Point-in-Time count – Hawai‘i (wildfire)	Wildfire	467 people added by the 2026 Point-in-Time count (increase over the likely demand: 16.2%)
141	23a	No car, inside the tsunami zone, over 15 minutes' walk to a safe zone – Hawai‘i	Tsunami	461 people without a car more than 15 minutes' walk from a tsunami safe zone (Census areas over 15 minutes' walk: 2; longest walk (minutes): 79.5)
142	14	Hearing difficulty and outside siren range (tsunami, at night) – Hawai‘i	Tsunami	455 people with hearing difficulty outside rated siren range (upper bound)
143	12	Power out 1 to 14 days – Hawai‘i (wildfire)	Wildfire	421 power-dependent residents in the risk area (outlets per day while the power is out: 421)
144	12	Power out 1 to 14 days – Hawai‘i (coastal flooding)	Coastal flooding	410 power-dependent residents in the risk area (outlets per day while the power is out: 410)
145	15	Power-dependent and far from a site – Hawai‘i (coastal flooding)	Coastal flooding	325 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 8)
146	15	Power-dependent and far from a site – Hawai‘i (wildfire)	Wildfire	290 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 8)
147	13	Kūpuna living alone, no car, far from a site – Hawai‘i (tsunami)	Tsunami	272 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 13)
148	15	Power-dependent and far from a site – Hawai‘i (inland flooding)	Inland flooding	182 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 4)
149	12	Power out 1 to 14 days – Hawai‘i (tsunami)	Tsunami	89 power-dependent residents in the risk area (outlets per day while the power is out: 89)
150	15	Power-dependent and far from a site – Hawai‘i (tsunami)	Tsunami	66 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 13)

Rank	Test	Scenario	Hazard	Result
325	17	Pet households vs pet-friendly sites – Hawai‘i (hurricane)	Hurricane	1,536 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 0)
326	17	Pet households vs pet-friendly sites – Hawai‘i (inland flooding)	Inland flooding	826 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 0)
327	17	Pet households vs pet-friendly sites – Hawai‘i (wildfire)	Wildfire	727 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 0)
328	17	Pet households vs pet-friendly sites – Hawai‘i (coastal flooding)	Coastal flooding	711 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 0)
329	17	Pet households vs pet-friendly sites – Hawai‘i (tsunami)	Tsunami	154 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 0)
330	19	Childcare centres in the hurricane zone – Hawai‘i	Hurricane	71 childcare centres in the zone (childcare centres in the county: 71)
331	19	Childcare centres in the wildfire zone – Hawai‘i	Wildfire	17 childcare centres in the zone (childcare centres in the county: 71)
332	19	Childcare centres in the tsunami zone – Hawai‘i	Tsunami	6 childcare centres in the zone (childcare centres in the county: 71)
333	19	Childcare centres in the coastal flooding zone – Hawai‘i	Coastal flooding	4 childcare centres in the zone (childcare centres in the county: 71)
334	19	Childcare centres in the inland flooding zone – Hawai‘i	Inland flooding	4 childcare centres in the zone (childcare centres in the county: 71)
335	22	Care facilities in the coastal flooding zone – Hawai‘i	Coastal flooding	1 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 1)
336	22	Care facilities in the inland flooding zone – Hawai‘i	Inland flooding	1 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 1)
337	16	Chuukese: 0 of 15 messages translated – Hawai‘i	All hazards	people who speak English less than very well: needs county data (messages translated: 0)
338	22	Care facilities in the tsunami zone – Hawai‘i	Tsunami	0 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 0)

Questions for counties

These need county knowledge. The plan has no data for them, so it shows no numbers.

1. **Shelter buildings in an earthquake or a strong hurricane:** Which shelter buildings have been assessed for earthquake and hurricane-wind resilience, and which need a check before they are used after a strong event?
2. **Resupply for Molokaʻi and Lānaʻi:** How would water, food, cots, fuel and medical supplies reach Molokaʻi and Lānaʻi if the harbours or airports close for several days, and who decides the order of shipments?
3. **Cell network outages:** Which areas lose mobile phone service first in a long power outage or after wind damage, and how will warnings and shelter information reach them (radio, sirens, door-to-door)?
4. **Schools used as shelters:** When a school is used as a shelter, how long can it stay open before classes must restart, and where do residents go next?
5. **Partner capacity:** Could each county collect partner capacity (cots, meals, staff, vehicles, interpreters) with a county-run form such as Survey123, with no personal data, so the plan can show pledged capacity instead of 'confirm with partner'?